

DUNMAN SECONDARY SCHOOL

CANDIDATE
NAME

CLASS

INDEX
NUMBER

PRELIMINARY EXAMINATION 2025
SECONDARY 4 EXPRESS / 5 NORMAL ACADEMIC

ADDITIONAL MATHEMATICS

Paper 1

4052/01

20 August 2025
2 hours 15 minutes

Solutions

Question	Answer
1	$x - 4 = \frac{2}{x - 3}$
	$(x - 4)(x - 3) = 2$
	$x^2 - 7x + 12 = 2$
	$x^2 - 7x + 10 = 0$
	$(x - 2)(x - 5) = 0$
	$x = 2 \text{ or } 5$
	When $x = 2, y = -2$
	When $x = 5, y = 1$
	Length = $\sqrt{(2 - 5)^2 + (-2 - 1)^2}$ = 4.24 unit ²
2a	$-90^\circ < \tan^{-1} p < 90^\circ \text{ or } -\frac{\pi}{2} < \tan^{-1} p < \frac{\pi}{2}$
2b	$\pi - x$
3	$3x^2 + 15x + 20 = 3(x^2 + 5x) + 20$
	$= 3[(x + 2.5)^2 - (2.5)^2] + 20$
	$= 3(x + 2.5)^2 + 1.25$
	Since $3(x + 2.5)^2 + 1.25 \geq 1.25$, $3x^2 + 15x + 20$ cannot be smaller than 1.

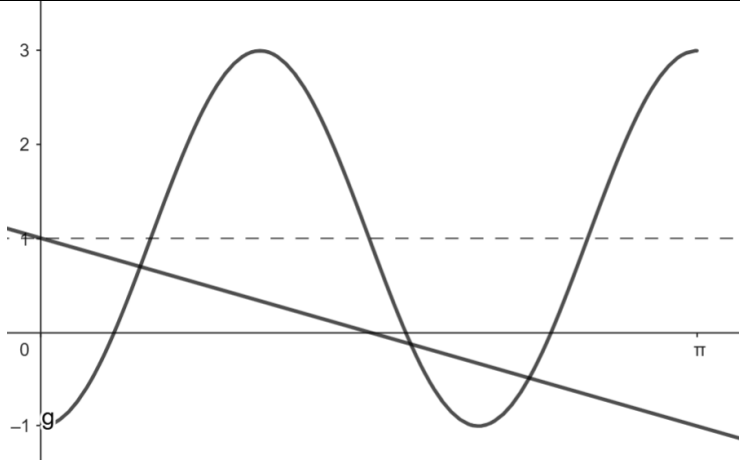
Question	Answer
4a	$\frac{x^2 + 3x + 2}{x^2(2x+1)} = \frac{A}{x} + \frac{B}{x^2} + \frac{C}{2x+1}$
	$x^2 + 3x + 2 = Ax(2x+1) + B(2x+1) + Cx^2$
	When $x = -\frac{1}{2}$, $\frac{1}{4} - \frac{3}{2} + 2 = C\left(\frac{1}{4}\right)$ $C = 3$
	When $x = 0$, $B = 2$
	Comparing coefficients of x^2 , $A = -1$
	$\frac{x^2 + 3x + 2}{x^2(2x+1)} = -\frac{1}{x} + \frac{2}{x^2} + \frac{3}{2x+1}$
4b	$\int \frac{x^2 + 3x + 2}{x^2(2x+1)} dx = \int -\frac{1}{x} + \frac{2}{x^2} + \frac{3}{2x+1} dx$ $= -\ln x - \frac{2}{x} + \frac{3}{2} \ln(2x+1) + c$
5	$3\log_5 y - \log_y 5 = 2$
	$3\log_5 y - \frac{\log_5 5}{\log_5 y} = 2$
	$3\log_5 y - \frac{1}{\log_5 y} = 2$
	$3(\log_5 y)^2 - 2\log_5 y - 1 = 0$
	$(3\log_5 y + 1)(\log_5 y - 1) = 0$
	$\log_5 y = 1$ or $-\frac{1}{3}$
	$y = 5$ or $y = 5^{-\frac{1}{3}} = 0.585$ (3s.f.)

Question	Answer
6a	$\sin(105^\circ) = \sin(60^\circ + 45^\circ)$
	$= \sin 60^\circ \cos 45^\circ + \cos 60^\circ \sin 45^\circ$
	$= \frac{\sqrt{3}}{2} \cdot \frac{1}{\sqrt{2}} + \frac{1}{2} \cdot \frac{1}{\sqrt{2}}$
	$= \frac{\sqrt{3}+1}{2\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}}$
	$= \frac{\sqrt{6}+\sqrt{2}}{4}$
6b	$\frac{1}{2}(4)(BC)\sin 105^\circ = 10 + 2\sqrt{3}$
	$\frac{1}{2}(4)(BC)\left(\frac{\sqrt{6}+\sqrt{2}}{4}\right) = 10 + 2\sqrt{3}$
	$BC = \frac{2(10+2\sqrt{3})}{\sqrt{6}+\sqrt{2}} \times \frac{\sqrt{6}-\sqrt{2}}{\sqrt{6}-\sqrt{2}}$
	$= \frac{4(5+\sqrt{3})(\sqrt{6}-\sqrt{2})}{6-2}$
	$= 5\sqrt{6} + \sqrt{18} - 5\sqrt{2} - \sqrt{6}$
	$= 4\sqrt{6} + 3\sqrt{2} - 5\sqrt{2}$
	$= 4\sqrt{6} - 2\sqrt{2}$
	$= \sqrt{2}(4\sqrt{3} - 2)$

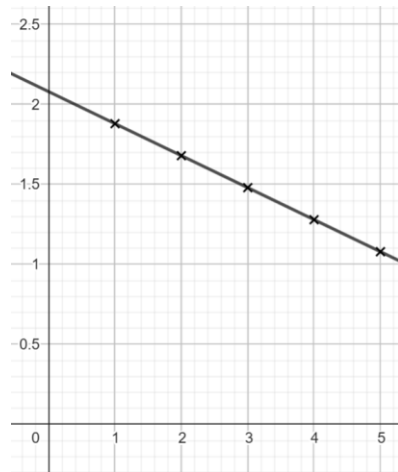
Question	Answer
7a	$9^x + 6^x = 6(4^x)$
	$\frac{9^x}{4^x} + \frac{6^x}{4^x} = 6$
	$\frac{3^{2x}}{2^{2x}} + \left(\frac{6}{4}\right)^x - 6 = 0$
	$\left(\frac{3}{2}\right)^{2x} + \left(\frac{3}{2}\right)^x - 6 = 0$
	When $u = \left(\frac{3}{2}\right)^x$, $u^2 + u - 6 = 0$
7b	$(u+3)(u-2) = 0$
	$u = 2$ or -3 (n.a.)
	$\left(\frac{3}{2}\right)^x = 2$
	$x = \frac{\ln 2}{\ln \frac{3}{2}}$ or $\frac{\lg 2}{\lg \frac{3}{2}} = 1.71$
8	When $x = -2$, $2(-2)^3 + a(-2)^2 + b(-2) + 10 = 0$
	$4a - 2b = 6$
	$2a - b = 3$ - eqn 1
	When $x = 3$, $2(3)^3 + a(3)^2 + b(3) + 10 = 10$
	$9a + 3b = -54$
	$3a + b = -18$ - eqn 2
	eqn 1 + eqn 2: $5a = -15$
	$a = -3$
	$\therefore b = -9$

Question	Answer
9a	$12t - \frac{3}{2}t^2 = 0$
	$t\left(12 - \frac{3}{2}t\right) = 0$
	$t = 0$ or 8
	$\therefore 8 \text{ s}$
9b	distance $= \int_0^8 12t - \frac{3}{2}t^2 \, dx$
	$= \left[6t^2 - \frac{1}{2}t^3 \right]_0^8$
	$= [128] - [0]$
	$= 128 \text{ m}$
9c	$a = 12 - 3t$
	When $t = 5$, $a = -3 \text{ m/s}^2$

Question	Answer
10	$y = xe^{-3x}$
	$\frac{dy}{dx} = e^{-3x}(1) + x(-3e^{-3x})$
	$= e^{-3x}(1 - 3x)$
	$e^{-3x}(1 - 3x) = 0$
	$x = \frac{1}{3}$
	$y = \frac{1}{3e}$ or 0.123
	$\frac{d^2y}{dx^2} = (1 - 3x)(-3e^{-3x}) + e^{-3x}(-3)$
	When $x = \frac{1}{3}$, $\frac{d^2y}{dx^2} < 0$
	$\left(\frac{1}{3}, -\frac{1}{3e}\right)$ is a maximum point.

Question	Answer
11a	 The graph shows two functions plotted on a coordinate system. The x-axis is labeled with 0 and π. The y-axis is labeled with -1, 0, 1, 2, and 3. A solid curve, $y = 2 \cos 3x$, starts at $(0, 2)$, reaches a minimum of $y = -2$ at $x = \frac{2\pi}{3}$, and returns to $y = 2$ at $x = \frac{4\pi}{3}$. A straight line, $y = -\frac{2}{\pi}x + 1$, starts at $(0, 1)$ and ends at $(\pi, -1)$. A dashed horizontal line is drawn at $y = 1$. The intersection points of the two graphs are marked with dots.
11b	The $2 \cos 3x = \frac{2}{\pi}x$ is equivalent to $-2 \cos 3x + 1 = -\frac{2}{\pi}x + 1$ and so its solutions can be found from the interception points of the two graphs.

Question	Answer
12a	Midpoint of $PQ = \left(\frac{1+9}{2}, \frac{-2+10}{2} \right) = (5, 4)$
	Gradient of $PQ = \frac{-2-10}{1-9} = \frac{3}{2}$
	Gradient of perpendicular bisector = $-\frac{2}{3}$
	$y - 4 = -\frac{2}{3}(x - 5)$
	$y = -\frac{2}{3}x + \frac{22}{3}$
12b	$-\frac{2}{3}x + \frac{22}{3} = 8x - 10$
	$-\frac{26}{3}x = -\frac{52}{3}$
	$x = 2$
	$R(2, 6)$
12c	Area = $\frac{1}{2} \begin{vmatrix} 9 & 2 & 1 & 9 \\ 10 & 6 & -2 & 10 \end{vmatrix}$
	$= \frac{1}{2} [(54 - 4 + 10) - (20 + 6 - 18)]$
	$= 26 \text{ unit}^2$

Question	Answer																		
13a	<table><tr><td>t</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>m</td><td>6.55</td><td>5.36</td><td>4.40</td><td>3.60</td><td>2.94</td></tr><tr><td>$\ln m$</td><td>1.88</td><td>1.68</td><td>1.48</td><td>1.28</td><td>1.08</td></tr></table>	t	1	2	3	4	5	m	6.55	5.36	4.40	3.60	2.94	$\ln m$	1.88	1.68	1.48	1.28	1.08
	t	1	2	3	4	5													
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$\ln m$	1.88	1.68	1.48	1.28	1.08														
																			
13b	$\ln M = 2.08$ $M = 8.00$																		
13c	Gradient = -0.2 $k = -0.2$																		

Question	Answer
14a	$\frac{d^2y}{dx^2} = -4\sin(2x + \pi)$
	$\frac{dy}{dx} = 2\cos(2x + \pi) + c_1$
	$3 = 2\cos\left[2\left(\frac{\pi}{2}\right) + \pi\right] + c_1$
	$3 = 2(1) + c_1$
	$c_1 = 1$
	$\frac{dy}{dx} = 2\cos(2x + \pi) + 1$
	$y = \sin(2x + \pi) + x + c_2$
	$\pi = \sin\left[2\left(\frac{\pi}{2}\right) + \pi\right] + \frac{\pi}{2} + c_2$
	$c_2 = \frac{\pi}{2}$
	$y = \sin(2x + \pi) + x + \frac{\pi}{2}$
14b	When $x = 3$, $\frac{dy}{dx} = 2\cos[2(3) + \pi] + 1 = -0.920$
	y is decreasing when $x = 3$.

Question	Answer
15	$y = \frac{16}{(x+1)^2} - 1 = 16(x+1)^{-2} - 1$
	$\frac{dy}{dx} = -32(x+1)^{-3}$
	$-4 = -32(x+1)^{-3}$
	$(x+1)^3 = 8$
	$x = 1$
	At L, $y = \frac{16}{(1+1)^2} - 1 = 3$
	$y - 3 = -4(x - 1)$
	$y = -4x + 7$
	At N, $0 = -4x + 7$
	$x = 1.75$
	At M, $0 = \frac{16}{(x+1)^2} - 1$
	$x = 3 \text{ or } -5 \text{ (n.a.)}$
	Area = $\left[\int_1^3 16(x+1)^{-2} - 1 \, dx \right] - \int_1^{1.75} -4x + 7 \, dx$
	$= \left[-16(x+1)^{-1} - x \right]_1^3 - \left[-2x^2 + 7x \right]_1^{1.75}$
	$= 2 - 1.125$
	$= 0.875 \text{ unit}^2$